

NUMBER SYSTEM

Type-1

1. Which of the following fraction is the smallest ? $9/13$, $17/26$, $28/29$, $33/52$
1) $33/52$ (2) $17/26$ (3) $9/13$ (4) $28/29$
2. Which of the following fraction is the smallest? $8/15$, $14/33$, $7/13$, $11/13$
(1) $8/15$ (2) $7/13$ (3) $11/13$ (4) $14/33$
3. Which of the following is the smallest fraction ? $8/25$, $7/23$, $11/23$, $14/53$
(1) $8/25$ (2) $7/23$ (3) $11/23$ (4) $14/53$
4. Which of the following is correct ?
(1) $2/3 < 3/5 < 11/15$ (2) $3/5 < 2/3 < 11/15$ (3) $11/15 < 3/5 < 2/3$ (4) $3/5 < 11/15 < 2/3$

Type-2

5. If $5432*7$ is divisible by 9, then the digit in place of * is :
(1) 0 (2) 1 (3) 6 (4) 9
6. If $78*3945$ is divisible by 11, where * is a digit, then * is equal to
(1) 1 (2) 0 (3) 3 (4) 5
7. A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232, etc. Any number of this form is always exactly divisible by :
(1) 7 (2) 11 (3) 13 (4) Smallest 3-digit prime number
8. A six digit number is formed by repeating a three digit number; for example, 256, 256 or 678, 678 etc. Any number of this form is always exactly divisible by:
(1) 7 only (2) 11 only (3) 13 only (4) 1001
9. When 335 is added to $5A7$, the result is $8B2$. $8B2$ is divisible by 3. What is the largest possible value of A ?
(1) 8 (2) 2 (3) 1 (4) 4
10. $(325 + 326 + 327 + 328)$ is divisible by
(1) 11 (2) 16 (3) 25 (4) 30

Type-3

11. The divisor is 25 times the quotient and 5 times the remainder. If the quotient is 16, the dividend is :
(1) 6400 (2) 6480 (3) 400 (4) 480
12. In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 46, then the dividend is
(1) 4236 (2) 430 (3) 4336 (4) 5336

NUMBER SYSTEM

13. In a division sum, the divisor is 3 times the quotient and 6 times the remainder. If the remainder is 2, then the dividend is

(1) 50 (2) 48 (3) 36 (4) 28

Type-4

14. Find the number of divisors or factor for 180?

1) 16 2) 20 3) 18 4) 14

15) Find the number of divisors or factors for 160?

1) 10 2) 12 3) 16 4) 14

16. Find the sum of divisors or factors for 120?

1) 360 2) 60 3) 240 4) 420

17) Find the sum of divisors of 100?

1) 175 2) 217 3) 240 4) 420

Type-5

18. Find the unit place value of $(573)^{678}$?

1) 9 2) 3 3) 6 4) 1

19. The digit in the unit's place of the product $(2464)^{1793} \times (615)^{317} \times (131)^{491}$ is

(1) 0 (2) 2 (3) 3 (4) 5

20. The digit in the unit's place of $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - 164 + 259]$ is :

(1) 1 (2) 4 (3) 5 (4) 6

21. The units digit of the expression $25^6 251 + 36^5 28 + 73^4 54$ is

(1) 6 (2) 5 (3) 4 (4) 0

Type-6

22. Find the number of zeros in the final product of $15 \times 25 \times 16 \times 9 \times 12 \times 75 \times 55$?

1) 3 2) 4 3) 6 4) 5

23. Find the number of zeros in the product of $60!$?

1) 12 2) 14 3) 10 4) 16

24. Find the number of zeros in the last product of $1 \times 2 \times 3 \times 4 \times \dots \times 98 \times 99 \times 100$?

1) 25 2) 24 3) 16 4) 14

NUMBER SYSTEM

Type-7

25.The total number of integers between 100 and 200, which are divisible by both 9 and 6, is :

(1) 5 (2) 6 (3) 7 (4) 8

26.How many numbers between 400 and 800 are divisible by 4, 5 and 6 ?

(1) 7 (2) 8 (3) 9 (4) 10

27.How many numbers between 1000 and 5000 are exactly divisible by 225 ?

(1)16 (2)18 (3)19 (4)12

TYPE-8

28.If 17200 is divided by 18, the remainder is—

(1) 17 (2) 16 (3) 1 (4) 2

29.The remainder when 321 is divided by 5 is

(1) 1 (2) 2 (3) 3 (4) 4

30. $96 - 11$ when divided by 8 would leave a remainder of :

(1) 0 (2) 1 (3) 2 (4) 3

31. $216 - 1$ is divisible by

(1) 11 (2) 13 (3) 17 (4) 19

TYPE-9

32.A number when divided by 49 leaves 32 as remainder. This number when divided by 7 will have the remainder as

(1) 4 (2) 3 (3) 2 (4) 5

33.When a number is divided by 36, the remainder is 19. What will be the remainder when the number is divided by 12 ?

(1) 7 (2) 5 (3) 3 (4) 0

34.When two numbers are separately divided by 33, the remainders are 21 and 28 respectively. If the sum of the two numbers is divided by 33, the remainder will be

(1) 10 (2) 12 (3) 14 (4) 16

NUMBER SYSTEM

TYPE -10

35. Two numbers, when divided by 17, leave remainders 13 and 11 respectively. If the sum of those two numbers is divided by 17, the remainder will be

(1) 13 (2) 11 (3) 7 (4) 4

36. Find the largest number of 4-digits which is exactly divisible by 23?

1) 9999 2) 9982 3) 9990 4) 9960

37. Find the largest number of 4-digits, when it is divisible by 23 leaves remainder 7?

1) 9989 2) 9982 3) 9990 4) None

38. Find the smallest 4-digit number which is exactly divisible by 37?

1) 1037 2) 1023 3) 1042 4) 1036

39. Find the least number of 4-digits which, when it is divided by 37 leaves remainder '3'?

1) 1036 2) 1039 3) 1037 4) 1033

TYPE-11

40. When a number is added to its $\frac{1}{4}$ it becomes 200. Find the number?

1) 180 2) 160 3) 120 4) 150

41. Weight of a bucket when filled fully with water is 17 kg. If the weight of the bucket when half filled with water is 13.5 kg, what is the weight of empty bucket?

(1) 12 kg (2) 8 kg (3) 10 kg (4) 7 kg

42. In a farm there are cows and hens. If heads are counted they are 180, if legs are counted they are 420. The number of cows in the farm is

(1) 130 (2) 150 (3) 50 (4) 30

43. I have x marbles. My elder brother has 3 more than mine, while my younger brother has 3 less than mine. If the total number of marbles is 15, the number of marbles that I have is

(1) 3 (2) 5 (3) 8 (4) 7

44. A man has some hens and some cows. If the total number of heads of hens and cows together is 50 and the number of feet of hens and cows together is 142, then the number of cows is

(1) 21 (2) 25 (3) 27 (4) 29

45. 0.1 and $\frac{5}{8}$ of a bamboo are in mud and water respectively and the rest of length 2.75 m is above water. What is the length of the bamboo?

(1) 10 m (2) 30 m (3) 27.5 m (4) 20 m

NUMBER SYSTEM

46. A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil was taken out and 4 bottles of oil was poured into it, it was $\frac{3}{4}$ full. How many bottles of oil can the tin contain ?

(1) 10 (2) 20 (3) 30 (4) 40

47. A man read $\frac{2}{5}$ th of a book on the first day. He read $\frac{1}{3}$ rd more on second day than he read on the first day. 15 pages were left for the third day. The number of pages in the book is

(1) 100 (2) 105 (3) 225 (4) 250

48. A tree increases annually by $\frac{1}{8}$ th of its height. By how much will it increase after 2 years, if it stands today 64 cm high?

(1) 72 cm (2) 74 cm (3) 75 cm (4) 81 cm

49. $\frac{1}{10}$ of a rod is colored red, $\frac{1}{20}$ orange, $\frac{1}{30}$ yellow, $\frac{1}{40}$ green, $\frac{1}{50}$ blue, $\frac{1}{60}$ black and the rest is violet. If the length of the violet portion of the rod is 12.08 meters, then the length of the rod is

(1) 16 m (2) 18 m (3) 20 m (4) 30 m

50. A, B, C and D purchase a gift worth 60. A pays $\frac{1}{2}$ of what others are paying, B pays $\frac{1}{3}$ of what others are paying and C pays $\frac{1}{4}$ of what others are paying. What is the amount paid by D ?

(1) 16 (2) 13 (3) 14 (4) 15